

PROJECT REPORT

Task 4: Password Generator (CLI)

Synent Technologies – Python Development Internship

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1. Introduction

The increasing use of digital platforms has made secure password creation an essential part of cybersecurity. Weak or predictable passwords increase the risk of unauthorized access and data breaches. This project focuses on developing a command-line application that generates secure and customizable passwords using Python. The project demonstrates modular programming, secure random generation techniques, and user-friendly interaction.

2. Objective

- Develop a secure password generator using Python.
- Allow users to customize password length and character sets.
- Evaluate password strength.
- Apply modular programming practices.
- Improve understanding of secure coding techniques.

3. Problem Statement

Many users create weak passwords that are easy to guess. This application helps generate strong passwords by combining different character sets using Python's secure random generation utilities.

4. Technologies Used

Programming Language: Python 3

Libraries: secrets, random, string

Development Environment: Visual Studio Code

Version Control: Git & GitHub

5. System Design

The application follows a modular architecture. The main module manages program execution, the menu module handles user interaction, the generator module creates passwords, the strength module evaluates password quality, and the utility module validates user inputs.

6. Project Workflow

1. Start the application.
2. Display the main menu.
3. Accept password preferences.
4. Validate user inputs.
5. Generate a secure password.
6. Evaluate password strength.
7. Display the generated password.
8. Repeat or exit.

7. Features Implemented

- Secure password generation using strong randomness.
- User-defined password length.
- Support for uppercase, lowercase, numeric, and special characters.
- Password strength evaluation.
- Modular folder structure.
- Input validation and exception handling.

8. Folder Structure

```
assets/  
docs/  
modules/  
    generator.py  
    menu.py  
    strength.py  
    utils.py  
main.py  
README.md  
requirements.txt
```

9. Testing and Validation

The application was tested with different password lengths and combinations of character types. Invalid inputs such as negative lengths, non-numeric values, and missing character selections were tested to verify validation logic. Generated passwords were manually inspected to confirm randomness and inclusion of the selected character sets.

10. Results

The application successfully generated secure passwords based on user preferences and provided immediate feedback regarding password strength. The modular implementation improved readability, maintainability, and future extensibility of the project.

11. Future Enhancements

- Copy generated password to clipboard.
- Export passwords to encrypted files.
- Password history management.
- GUI version using Tkinter.
- Web-based version using Flask.

12. Conclusion

This project successfully achieved its objective of creating a secure and customizable password generator. It strengthened understanding of Python programming, modular software design, input validation, and secure password generation techniques while providing a practical utility that can be extended in future.

Screenshots

Main Menu:

```
=====
                        SECUREPASS CLI
=====
                Professional Password Generator
=====

Main Menu
-----
1. Generate Password
2. About
3. Exit

Enter your choice:
```

Password Generation:

```
Choose Password Mode
-----
1. Recommended (12 Characters)
2. Strong (16 Characters)
3. Ultra Secure (20 Characters)
4. Custom Length

Select Password Mode: 3

Select Character Types
Include Uppercase Letters? (Y/N): Y
Include Lowercase Letters? (Y/N): Y
Include Numbers? (Y/N): Y
Include Special Characters? (Y/N): Y

=====
                        GENERATED PASSWORD
=====

Password : fa2+^yYxuLd,0^?D?6@7

=====
Length           : 20
Strength         : Ultra Strong
Generated At     : 14-Jul-2026 08:25:16 AM
Uppercase Letters : Yes
Lowercase Letters : Yes
Numbers         : Yes
Special Characters : Yes
=====
```